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Enhancing understanding of vulnerability to climate change in rural territories of Tunisia: from local perceptions to a quantitative assessment

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Climate change represents today one of the most pressing environmental and socio-economic challenges facing humanity. However, its effect is uneven across regions, affecting most severely those whose populations depend directly on natural resources for their livelihood. Many studies have assessed climate vulnerability at national and regional levels, but there remains a lack of detailed analysis of how vulnerability manifests in rural areas and how it varies across local contexts, particularly in rural territories such as those in Tunisia. To address this gap, we investigate the vulnerability of local population to climate change in Rihana, a small rural territory in Sidi Bouzid governorate (Central Tunisia), through two complementary approaches. The first is a qualitative approach based on a participatory territorial diagnosis to assess the perception of climate change, current and future vulnerability; while the second is a quantitative approach based on the Climate Change vulnerability index_Intergovernmental Panel on Climate Change (LVI-IPCC). This methodology involves a comparative study of two Homogeneous Territorial Units (HTU) located upstream (HTU2) and downstream (HTU1) of the study area. The use of both qualitative and quantitative approaches provides a more comprehensive understanding of vulnerability, unlike other studies that limit themselves to a single approach and may therefore overlook certain crucial aspects. The results reveal that the local population in the study area perceives climate change as a threat to its main resources, particularly changes in rainfall and temperature patterns, drought and floods. Moreover, vulnerability levels vary significantly between the two areas, mainly due to differences in geographical position. The upstream part is more vulnerable (LVI-IPCC = 0.14), being more sensitive, more exposed and with a lower capacity to adapt, while the downstream part is less vulnerable (LVI-IPCC = 0.03). The results also show that households anticipate an increase in future vulnerability, with intensified soil erosion, declining rain-fed crops, growing water scarcity, and a heightened risk of rural exodus. Finally, this study confirms that even at small local scale vulnerability levels vary significantly so it's important to integrate local knowledge with scientific tools when analyzing vulnerability at the local level. Based on these findings, the study emphasizes the use of a territorial approach and the active involvement of local communities. Such engagement enhances the resilience of rural populations and facilitates the development of effective adaptation strategies, while providing valuable insights for future research on climate.

KEYWORDS
climate change, local scale, participatory territorial diagnosis, LVI-IPCC, rural livelihoods, Tunisia, vulnerability assessment

1 Introduction

Climate change represents one of the most pressing challenges facing humanity today. It is a very real phenomenon with significant impacts on ecosystems, societies and economies (Haden et al., 2012; IPCC, 2023). According to the Intergovernmental Panel on Climate Change (IPCC), human activities have caused global warming of about 1.1 °C above pre-industrial levels, for the period from 2011 to the 2020. This warming could reach 1.5 °C between 2030 and 2050 if it continues to increase at the same rate (IPCC, 2023). With such warming, climate change is expected to reduce agricultural productivity, threatening food security, and people's livelihoods. Global climate change will also increase the occurrence of drought, floods, and other climate extremes (USAID, 2018; IPCC, 2023). However, these effects do not equally affect natural and human systems across the world (Touzard and Ollat, 2021; IPCC, 2023). Feindouno (2022) points out that one of the many injustices of climate change is that it affects developing countries more severely. Particularly affected, Tunisia ranks among the countries most vulnerable to the impacts of climate change (Mahdhi et al., 2019; Ben Nasr et al., 2021). Over the past three decades, Tunisia has experienced a significant increase in temperatures, with a rise of 0.37 °C per decade. At the same time, there has been a decrease of about 3% in precipitation (World Bank, 2021). Climate change has exacerbated an already difficult situation in Tunisia, characterized mainly by scarce water resources. Its water potential remains limited to 4.8 billion m^3 per year (less than 420 m^3/year/capita). This situation places Tunisia in a state of structural water stress, as these supplies, which are decreasing year on year, are below the threshold of 500 m^3/inhabitant/year, widely recognized as the threshold for severe water stress (MARHP, 2021). Thus, climate change is a real and pressing challenge in Tunisia.

Rural territories of Tunisia, which account for nearly 34% of the country, are one of the most vulnerable regions to the impacts of climate change (Ben Nasr et al., 2021), yet they have received a little attention in research and policy discussions. Climate change poses environmental, social, and economic challenges in these regions because of their high dependency on climate-sensitive natural resources, dependence on rainfed agriculture, and lack of resources that limits their ability to cope with and adapt to various stresses (Chebbi et al., 2019; Zaidi et al., 2023). These vulnerabilities are further intensified by their geographic context, as they are often located in marginal areas and more exposed to climate-related risks, and by generally poor socioeconomic conditions (Mahdhi et al., 2019; Kadri et al., 2023). The consequences of climate change on Tunisian agricultural production systems would then be a change in land use, significant declines in crop yields, and the disappearance of some rainfed crops. According to projections, there will be a 25% reduction in cereal areas by 2030, which will result in a 40% decline in rainfed cereal production, and a 52% decrease in olive oil production (CIHEAM, 2016; Ben Nasr et al., 2021). Several studies also highlight that climate change increases the vulnerability of Tunisian rural livelihoods whose adaptive capacities are generally limited, making them more exposed to risks as droughts and floods (Mahdhi et al., 2019; Aribi and Sghaier, 2021; Zaidi et al., 2023).

A clear understanding of the impacts and the vulnerability of Tunisian rural territories to climate change is key to strengthening the resilience of rural communities and to designing effective adaptation strategies. Indeed, vulnerability assessments have become necessary given the uncertainty of climate change impacts at the local level (Zahnow et al., 2025). Although climate change is a global phenomenon, vulnerability is highly site-specific (Oo et al., 2018). It is therefore very important to understand what is happening at the local level (Choudri et al., 2013). Even within the same region, rural communities may experience different levels of vulnerability due to climatic and geographical heterogeneity and differences in the factors that influence a system or people's capacity to adapt (Dumenu and Takam Tiamgne, 2020). Consequently, not all communities are equally vulnerable (Parsons et al., 2024; Zahnow et al., 2025).

The official emergence of the concept of vulnerability in climate change science dates back to the IPCC's third report published in 2001. Since then, this concept has remained central to the IPCC's studies, also appearing in its latest report (Pandey et al., 2017; IPCC, 2022). In its Third Assessment Report, the IPCC (2001) defines vulnerability as "the degree to which a system is susceptible or unable to cope with the adverse effects of climate change, including climate variability and extremes. Vulnerability is thus often considered to be an interdependent function of exposure, sensitivity and adaptive capacity". In this definition, exposure refers to the rate and magnitude of climate change. Sensitivity corresponds to 'the degree to which a system is influenced, positively or negatively, by climate-related stimuli'. Adaptive capacity, on the other hand, refers to 'the set of capabilities, resources and institutions of a country or region that enable it to implement effective adaptation measures' (IPCC, 2001; Hassan and Kumar, 2019; Garba et al., 2022).

Over the past decades, various methods have been developed to assess vulnerability. Those methods adopt various approaches to systematically examine and integrate interactions between humans and their physical and social environments (Shah et al., 2013). A bottom-up approach where the unit of analysis is generally smaller and more localized, such as households or communities. This approach draws on the perspectives and knowledge of communities to understand current and future vulnerabilities. Local knowledge is often integrated through participatory processes. This approach therefore uses a qualitative method for vulnerability assessment (Nelitz et al., 2013). Another top-down approach uses indicators to characterise and quantify vulnerability, often combining various indicators into a single composite vulnerability index. These assessments have addressed different spatial scales (country, region, municipality, community, etc.) (Cutter et al., 2003). These indices have been classified into various groups, such as: the Social Vulnerability Index (SoVI) (Cutter et al., 2003), the Livelihood Vulnerability Index (in its two models: LVI, LVI-IPCC) (Hahn et al., 2009), and the Climate Vulnerability Index (CVI) (Pandey et al., 2017), etc.

These indices make it possible to compare communities or regions spatially. However, vulnerability is not only a measurable outcome but also a matter of perception (Adeoti et al., 2016). Households'

perceptions of climate change are influenced by their understanding, beliefs, and experiences about climate change issues (Widayati et al., 2021). For farming communities, to implement effective adaptation measures, they need to be aware of the issues about climate change. People who underestimate or ignore these impacts are often less inclined to adapt and manage the consequences. Thus, local perceptions of climate change play a crucial role in determing the way rural communities adapt to climate change (Wang et al., 2021; Widayati et al., 2021; Gemeda et al., 2023). Therefore, considering both indices and perception-based approaches thus provides a more nuanced and realistic picture of vulnerability.

Despite Tunisia's increasing exposure to climate change, there remains a lack of detailed analysis of how vulnerability manifests in rural areas and how it varies across local contexts. Existing studies often focus on national or regional scales, which may overlook the specific challenges faced by rural communities (Mahdhi et al., 2019). Furthermore, most studies on vulnerability to climate change in Tunisia focus mainly on quantitative assessment and, often overlooking how communities perceive climate change impacts and vulnerability, adaptive capacity, and lived experiences (Ben Nasr et al., 2021; Aribi and Sghaier, 2021). As a result, there is a critical gap in the literature regarding integrated, locally grounded, and perception-based assessments of rural climate vulnerability. In this context this study therefore seeks to fill this gap. It calls for an analysis of the vulnerability of the local population's livelihoods to climate change using two complementary methods: one qualitative, to see how the local population perceives climate change using a participatory approach, and the other quantitative, based on the Livelihoods Vulnerability Index (LVI-IPCC). The LVI-IPCC has been widely used by many researchers around the world due to its flexibility (Hahn et al., 2009; Mudasser et al., 2020). This index is applied in two homogeneous territorial units (HTUs) in the rural area of Rihana, in the governorate of Sidi Bouzid. HTU2 represents the mountainous area (upstream) while HTU1 represents the plain (downstream). Specifically, this study addresses the following four objectives: (i) to understand how climate change manifests at the local level, (ii) to assess the local perceptions on climate change and vulnerability, (iii) to identify the key factors that shape local vulnerability, and to measure the vulnerability of local livelihoods using the LVI-IPCC index.

2 Materials and methods

2.1 Study area

Rihana is one of the target sites of the PACTE program (Programme d'Adaptation aux Changements Climatiques des Territoires vulnérables de Tunisie -PACTE).[1] It is located in the Regueb delegation, Sidi Bouzid governorate (Central Tunisia) (Figure 1). The selected territory covers approximately 10,000 ha, 40% of which is private agricultural land and 60% is grazing land (Figure 2). Rihana is characterized by diverse landscapes. HTU2, which represents the upstream part of Rihana, consists of scrubland and nature olive, rangeland, and rainfed olive, However, HTU1 (the downstream plain) is occupied by irrigated crops, which covered approximately 6.4% of the total area, particularly olive trees and citrus (Figures 1, 2). Rihana agriculture is characterized by small farms, usually smaller than 10 ha with low economic productivity (Kadri et al., 2023).

The study area is located in the lower semi-arid climate zone, where the annual average temperature is around 20 °C, with a summer temperature reaching up to 47 °C (Figure 3B). It is characterized by irregular and fairly low rainfall: 225 ± 129 mm (Figure 3A). The study area was selected due to its higher exposure to climate change impacts. Rihana is one of the most economically and environmentally vulnerable areas of Tunisia (Gasmi et al., 2022). The vulnerability of natural resources, including soil degradation, water scarcity, and erosion, is complicated by the temporal and spatial variability of rainfall, which results in alternating periods of drought and flooding. The risk of flooding is high in Sidi Bouzid. Stormy rains cause significant surface runoff, resulting in flooding in almost all of the region's oueds. Sidi Bouzid and Rihana in particular have been exposed to several exceptional floods, such as those of 1969 and 1990 (Hajjem, 1999). Rainfall data analysis from 1980 to 2021 reveals several annual extremes, such as the wet year 2013–2014 (395.5 mm). About 10 deficit years, with a minimum of 50 mm for the year 2000–2001. The average years remain dominant (Figure 3A). The annual rainfall depth, annual average, mean, maximum, and minimum temperatures are shown in Figures 3A,B.

2.2 Qualitative method

The study employed a participatory territorial diagnosis to investigate households' perceptions of climate change. It serves to engage the local population of Rihana in an open dialogue on climate change and vulnerability issues. Our objective is to see how climate change is perceived in Rihana from the perspective of the local population and to assess its impact, thereby enabling us to analyse the vulnerability of this population's livelihoods. The approach took into account the different douars in Rihana with the participation of diverse social groups (men, women, young people and the elderly). A purposive sampling approach was used to select 52 households distributed across the entire Rihana territory, covering upstream, and downstream areas, in order to ensure homogeneous territorial representation. This approach took place in June 2020. The sample was balanced between the two HTUs, with 26 households from the upstream HTU (HTU2) and 26 households from the downstream HTU (HTU1), reflecting our aim from the outset of the study to include an equal number of participants in each unit.

Three complementary tools were used: collective discussions (I), participatory mapping (II), and participatory observations (III). The results were then shared with the local population for validation and potential adjustment. They were presented in three stages: first to the population of HTU1, then to the population of HTU2, and finally to both groups together to broaden the discussion.

(I) Collective discussions aimed to explore participants' perceptions and experiences of climate change. These discussions were conducted using four predefined categories related to climate change: beliefs, causes, consequences, and effects. Participants were invited to express their perceptions within each category. The information collected was then summarized using a descriptive analysis based on the frequency and percentage distribution of responses among each unit. This step offered a common community perspective, by emphasizing both biophysical risks and socioeconomic fragilities.

1 https://www.cirad.fr/dans-le-monde/cirad-dans-le-monde/projets/projet-pacte

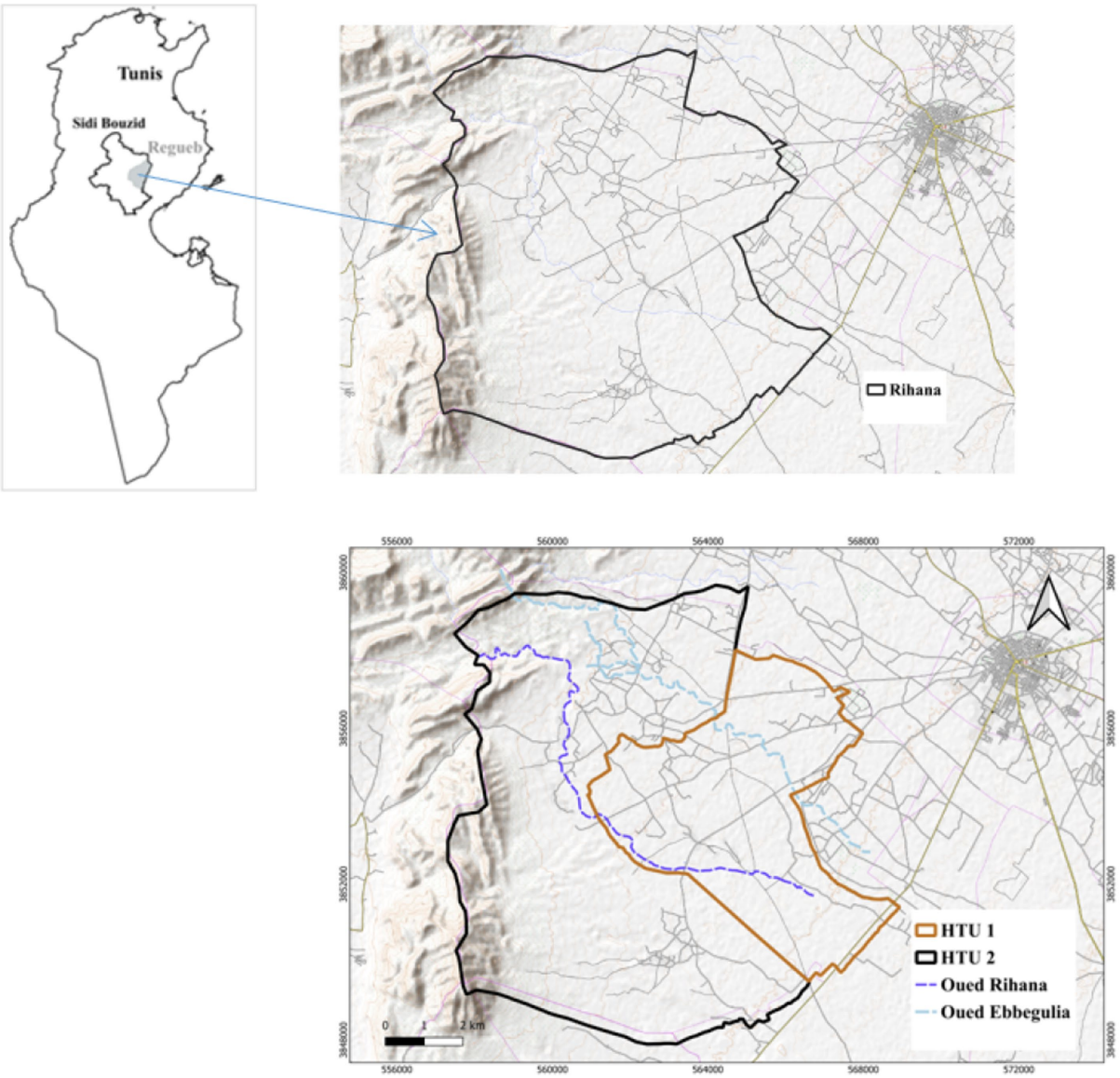

FIGURE 1
Geographic location of the Rihana study area in Tunisia.

(II) Participatory mapping involved engaging participants to locate and map the resources that were most significant to them on large sheets of paper, while explaining how these resources are exposed to climate risks, and their concerns about their future vulnerability. The resulting maps and the accompanying narratives were then synthesized and incorporated, thereby complementing the information obtained from collective discussions. This approach enabled us to create spoken maps. According to Burte (2018), 'carte parlée' is a map that represents the territory of life from the perspective of the inhabitants of that territory. It is drawn by local actors during group or individual interviews. It is an interactive tool that facilitates understanding of the different components of the territory and their specific organization.

(III) Participatory observations were carried out the local population of Rihana in order to relate the perceptions shared during collective discussions to concrete elements of the territory. At this step, residents were accompanied in their environment so they could point out areas and situations they perceived vulnerable to climate change. This method offered an objective perspective to the analysis of local vulnerability.

2.3 Quantitative method

The Livelihood Vulnerability framework is especially relevant to understand vulnerability to climate change since it offers a framework for analyzing both the main elements that make up livelihoods and the contextual factors that influence them (Hahn et al., 2009). The Livelihood Vulnerability Index (LVI) was developed based on the Sustainable Livelihoods Approach (SLA) and the work of Hahn et al. (2009). The SLA approach is a conceptual tool used to enhance understanding of the livelihoods of the low-income household. It examines five types of household assets: natural, social, financial, physical and human capital (Chambers and Conway, 1992). The LVI is presented in two approaches, as proposed by Hahn et al. (2009): the first expresses

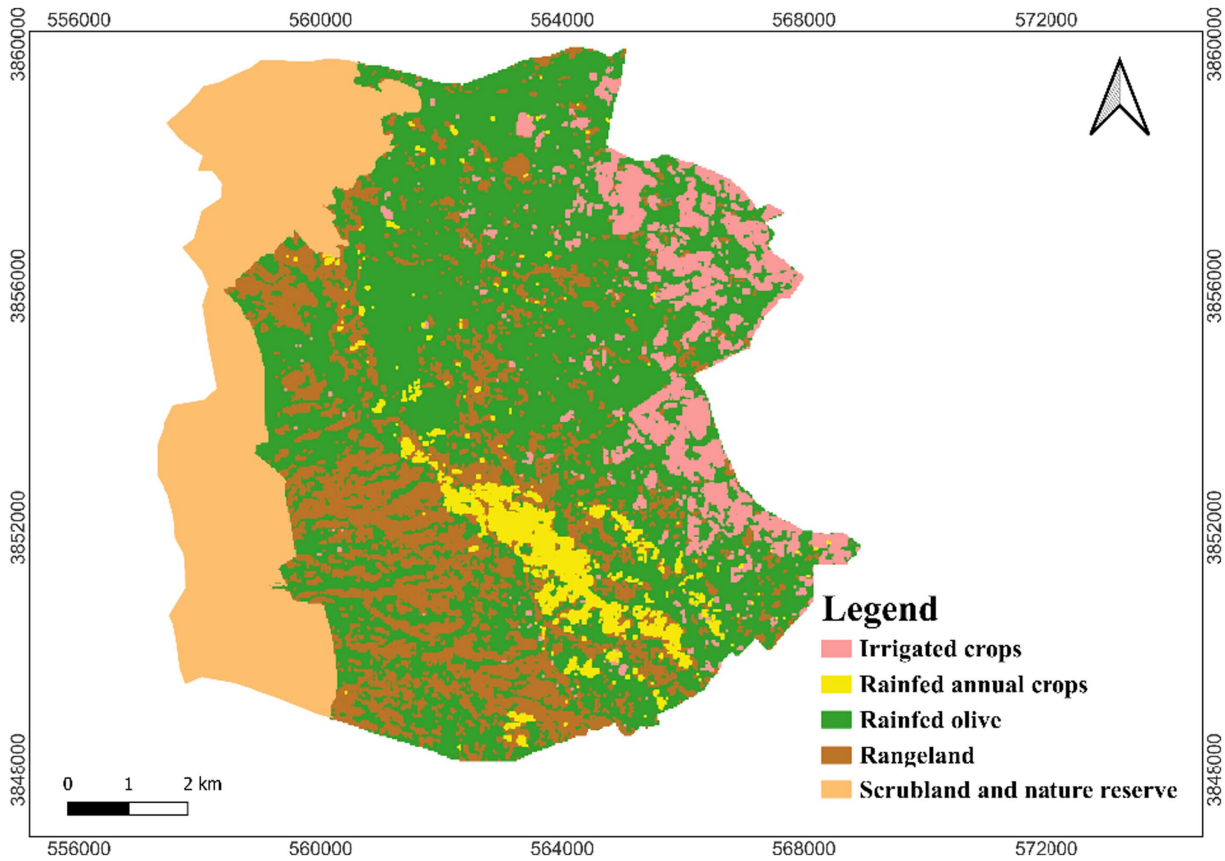

FIGURE 2
Land use land cover map of Rihana for the year 2020 (Kadri et al., 2023).

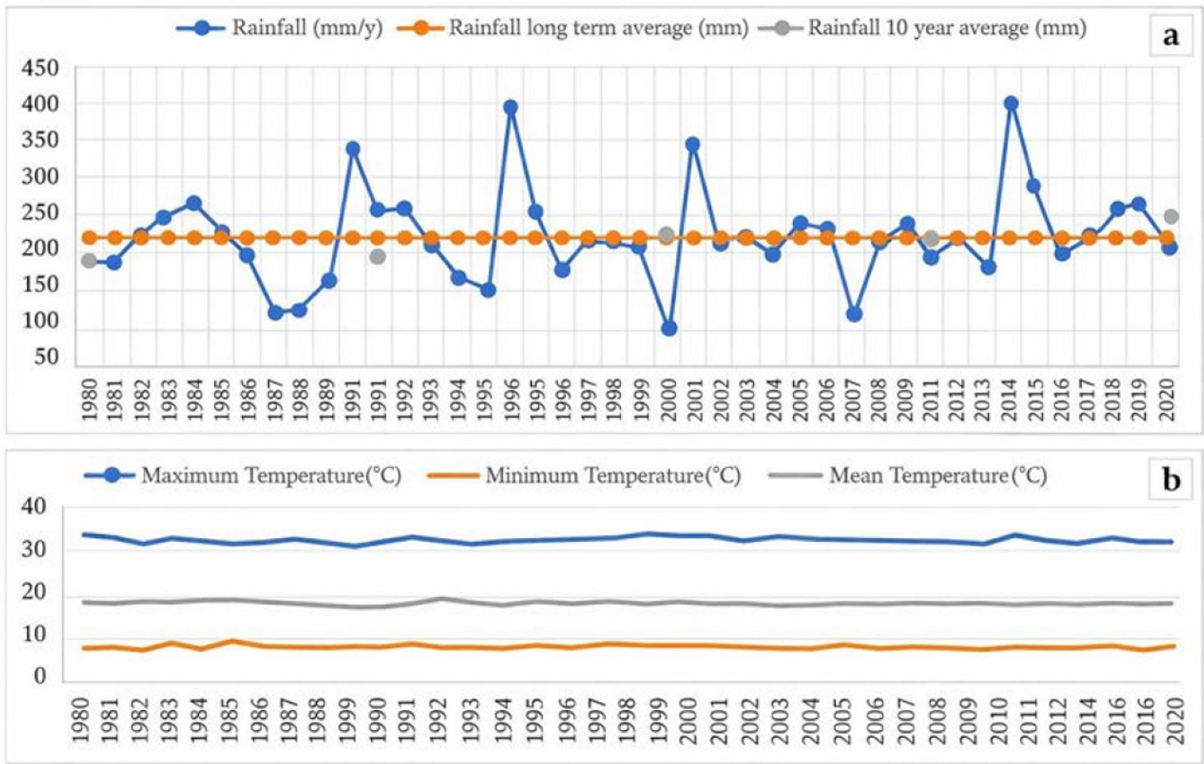

FIGURE 3
(a) Average annual rainfall; **(b)** Temperature (mean, maximum, and minimum) observed at Rihana.

the LVI in seven main components, while the second, LVI-IPCC, aggregates the seven into three factors (adaptive capacity, sensitivity and exposure) of IPCC vulnerability. In this case, adaptive capacity is defined by the contribution of the components "sociodemographic profile," "livelihood strategies" and "social networks." Sensitivity is defined by the components "food security," "water" and "health." Exposure is a factor linked mainly to natural disasters and climate variability. Each component is made up of several indicators or sub-components.

In our research, we adapted the LVI-IPCC to the semi-arid Tunisian context based on collective discussions with the local population of Rihana (26 HTU2, 26 HTU1), held before the survey. The local population's perceptions and experiences of climate extremes helped identify the factors that limit their ability to adapt to climate change. The resulting recommendations led to specific modifications to the LVI-IPCC index. Five main components were added to the seven components suggested by Hahn et al. (2009). These components are as follows: infrastructure, as a means to capture adaptive capacity, as well as natural resources, human resources, water availability and access, and land tenure, which are used to ultimately assess household sensitivity to climate change. According to local households, fragile infrastructure and limited connectivity increase vulnerability by reducing adaptive capacity. They requested the inclusion of natural resources, as their degradation and loss heighten vulnerability to climate-related shocks and threaten the basis of agriculture, livestock, and other livelihoods. Similarly, households emphasized the importance of human resources, as limitations in local labor availability increase sensitivity to climate-related shocks. They also requested the inclusion of the water availability and access component, as multiple concerns related to water resources were identified, and climate change has further increased the vulnerability of these resources. Finally, households highlighted the importance of including land tenure, recognizing that insecure rights and small, fragmented farms contribute to increased vulnerability. Alongside these components, new sub-components were then added to all the main components, and sub-components that were not relevant in our context were removed. Table 1 resumed the finalized LVI-IPCC with all the main components used.

2.4 Data collection

Individual surveys were conducted to collect the information needed to calculate the LVI-IPCC. The survey involved 52 households, almost the same participants who had previously taken part in collective discussions to assess perceptions of climate change (26 from HTU1 and 26 from HTU2). These households were re-contacted using the participant lists established during the participatory phase, with the support of the local territorial community involved in the PACTE program, which facilitated the maintenance of contact and the re-mobilization of the households, ensuring continuity between the qualitative and quantitative data. For each survey, we sought to collect information about the household, their family and their farm (age, marital status, level of education, main activity, family size, number of households, number of working people, head of the family, total area, number of plots, different crops, livestock, extra-agricultural activity, etc.), human and natural resources, health, food security and local strategies for adapting to climate change, etc. Our survey was conducted between June and September 2021.

TABLE 1 The main components constituting the three contributing factors of vulnerability.

Factors	Main component
Adaptive capacity	Sociodemographic profile
	Livelihood strategies
	Social Networks
	Infrastructure
Sensitivity	Health
	Food
	Water availability and access
	Natural resources
	Human resources
	Land tenure
Exposure	Climate variability and change

Bold values represent values added by the population.

2.5 Data analysis

The calculation of LVI-IPCC relies on a weighted average approach (Hahn et al., 2009). Each sub-component was considered to contribute equally to the overall index, even though each major component contains of different numbers of sub-components. A simple method with equal weights was therefore applied for all major components. Because the sub-component was measured on a specific scale, they were normalized as an index. In developing this index, Hahn et al. (2009) opted for equal weighting, a choice intended to create a simple and accessible assessment tool that can be readily adapted to different contexts. This approach has since been adopted by several other authors worldwide, and in Tunisia specifically by Aribi and Sghaier (2021) and by Fetoui et al. (2021).

For this purpose, the equation used to calculate the LVI-IPCC was applied (Equation 1):

$$\text{Index S} = \frac{(\text{S} - \text{Smin})}{(\text{Smax} - \text{Smin})} \tag{1}$$

Where S is the sub-component observed at the territory level and Smin and Smax are the minimum and maximum values for each sub-component.

However, recent studies have argued that Equation 1 should be used only when the sub-component has a positive relationship with the vulnerability (Adu et al., 2018; Zhang et al., 2019). Otherwise, they suggested Equation 2.

$$\text{Index S} = \frac{(\text{Smax} - \text{S})}{(\text{Smax} - \text{Smin})} \tag{2}$$

In step 2, the average standardised scores for each principal component are calculated using Equation 3, giving a final score (M) for each principal component.

$$M = \frac{\sum_{i}^{n} \text{Index Si}}{\text{n}} \tag{3}$$

Where n is the number of subcomponents in each component.

Once the values for each of the main components have been calculated, we move on to step 3, which involves calculating the three vulnerability factors according to Equation 4.

$$CF = \frac{\sum_{i}^{n} W_{Mi} \ Mi}{\sum_{i}^{n} W_{Mi}} \quad (4)$$

Where CF denotes the three vulnerability factors, W_{Mi} describes the weights of the principal components corresponding to the number of sub-components for each principal component, Mi is the principal component indexed by *i*, and *n* is the number of principal components in each factor. Finally, the LVI-IPCC is calculated by combining the three vulnerability factors using the following Equation 5:

$$LVI - IPCC = (e - a) * s \quad (5)$$

The values e, *a* and *s* are, respectively, those of exposure, adaptive capacity and sensitivity calculated.

The LVI-IPCC is scaled from −1 (less vulnerable) to +1 (more vulnerable) (Hahn et al., 2009).

3 Results

3.1 Local perception of climate change and vulnerability

During the collective discussions with the 52 rural households of Rihana (26 in HTU2 and 26 in HTU1), most of the participants believed that climate is no longer the same as it was in the past, it's changing. They agreed that 'climate change is a very serious issue'. Households understood climate change through different indicators. These indicators were split into four dimensions, namely belief, causes, consequences, and effects that were explored by different statements (Table 2). These collective perceptions, enriched by concrete experiences and shared narrative. HTU2 households (about 90%) often explained these changes as part of natural cycles. However, HTU1 households (nearly 40%) mentioned human activities such as the overuse of water for irrigation, and in general the poor agricultural practices, showing a stronger attribution to human influence.

Regarding the consequences, both HTU2 and HTU1 participants strongly highlighted warmer summer, cold winter, change in rainfall pattern, change in temperature pattern, more frequent and prolonged drought, and more violent and sudden floods as major consequences of climate change. They also observed strong wind as a consequence, but it is not considered as important as the other climate-related changes (Table 2). The increasing frequency of drought periods was particularly emphasized by all participants, who also cited specific years of extreme drought (1987, 2000, 2007, 2010, 2014–2017, 2019, 2020). Floods were noted mainly by HTU2 households (76%), compared to 34% in HTU1 areas (1990, 1996, 2020).

Regarding the effects of climate change (Table 2), all HTU2 participants perceived several key impacts, including the widening and deepening of the oueds (Oued Rihana and Oued Ebbegulia, figure1), damage to water and soil conservation techniques-which local communities heavily rely on to collect runoff and protect against floods-and the abandonment of certain agricultural lands. However, these three effects were considered less important by HTU1 participants, with only 23 and 15% mentioning them, respectively. HTU2 participants also emphasized production losses, livestock decline, loss of fertility and soil erosion, and water scarcity reflecting the steep slopes and mountainous conditions of their area (HTU2). In contrast, downstream participants (HTU1) recognized the same impacts but with lower frequency. During the participatory observation, they particularly highlighted the depletion of Oueds, noting that because upstream communities extensively use water and soil conservation techniques,

TABLE 2 Household perceptions of climate change (%).

Climate change perception	HTU2	HTU1
a. Belief		
Climate is changing	98	100
b. Cause		
Nature	90	60
Human influence	10	40
c. Consequence		
Warmer summer	46	37
Cold winter	57	22
Change in rainfall pattern	100	100
Change in temperature pattern	85	74
More frequent and prolonged drought	100	92
More violent and sudden floods	76	34
Strong winds	23	19
d. Effect		
Production loss	54	23
loss of soil fertility and soil erosion	76	31
Water scarcity	42	39
Depletion of Oueds	31	85
Widening and deepening of the Oueds	100	23
Loss of livestock	42	33
Abandonment of certain agricultural land	53	23
Damage to roads/houses	61	23
Damage to water and soil conservation techniques	100	15
Loss of agricultural income	35	27
Water-related conflicts	42	31
Loss of motivation to pursue farming	23	12
Rural exodus	31	11

water from Oueds-especially Oued Rihana-rarely reaches the downstream areas, contributing to the depletion of these oueds.

For HTU2 participants, participatory mapping enabled households to express their perceptions of current and future vulnerability to climate change through their own spatial representations. Participants began by drawing the mountainous area, grazeland area, and the two Oueds (Rihana and Ebbegulia). They then represented the main local agricultural resources, including rain-fed olive cultivation, cereal farming, and sheep and cattle breeding, which are considered the pillars of their rural economy. The maps produced highlight severe soil degradation due to erosion, perceived as one of the major impacts of climate change, as well as the destruction of water and soil conservation techniques caused by increasingly intense flooding. In terms of future vulnerability, households anticipate a worsening of these phenomena: soil erosion is expected to intensify, leading to the increasing destruction of soil and water conservation techniques, while rain-fed crops, particularly olives and cereals, are expected to decline in area and yield. In addition, they anticipate a scarcity of water resources, as the Rihana Oued will no longer supply the douars it once supplied. These representations reflect a local awareness of increasing vulnerability, marked by the degradation of natural resources (Table 3).

In the second area, located downstream (HTU1), participants indicated that agriculture there relies mainly on irrigated crops, although some households still practice rain-fed olive and cereal farming. Soil erosion is also present, but remains less pronounced than in the first area. With regard to future vulnerability, households believe that rain-fed farming will become increasingly difficult due to the expected decline in rainfall and soil degradation. They fear that they will be forced to rely more on irrigation, an option that is not available to all households due to a lack of sufficient financial and material resources. This growing dependence on irrigation, combined with inequalities in access to water, could exacerbate the precarious situation of some rural families. Participants also expressed concern about an increased risk of rural exodus, perceived as a direct consequence of declining agricultural opportunities and the gradual loss of livelihoods.

Finally, conclusions drawn from local perceptions of vulnerability reveal that this population is fully aware of the current and future impacts of climate change on their territory, particularly with regard to variations in temperature, rainfall, flooding and periods of drought, as well as erosion. The variations in rainfall and temperatures mentioned by the local population correspond remarkably well with those identified in the statistical analysis of climate variables (rainfall, temperature; Figure 3).

3.2 Assessment of vulnerability using the LVI-IPCC index

The findings indicate that the local population of Rihana in both HTUs are vulnerable to the impacts of climate change. However, the LVI-IPCC index value for HTU2, which is 0.14, is higher than that for HTU1, which is 0.03. This finding shows that HTU2 is characterized by high levels of sensitivity and exposure, together with a low adaptive capacity (Figures 4, 5).

3.2.1 Assessment of adaptive capacity

In terms of adaptive capacity, HTU2 appears to be more vulnerable than HTU1 due to its low adaptive capacity to climate change (0.35 compared to 0.44 in HTU1). This is mainly explained by a higher proportion of households reported depending exclusively on agriculture as their main source of income (98% compared to 76% in HTU1) and the predominance of rain-fed farming, which makes production highly (87% compared to 44% in HTU1). A larger share of households in HTU1 have access to irrigation, allowing them to better cope with water scarcity and maintain stable production. Furthermore, the low adaptive capacity of HTU2 is explained through the "infrastructure" component, with values of 0.42 at HTU2 compared to 0.26 at HTU1. This difference can be explained by the rugged nature of the terrain. Hence the lack of investment capable of preventing the population from becoming isolated during floods. However, the situation is easier in HTU1, which is located on the plain and is better served by tracks, roads and other land development. Moreover, the "Social Network" component has a greater influence on the adaptive capacity of the population of Rihana (0.65 compared to 0.4 in HTU2). A larger share of households in the HTU2 area do not belong to any organization or association, suggesting lower social cohesion and reduced access to support networks. Similarly, fewer households in the HTU1 area benefit from assistance provided by friends, associations, communities, or the state.

3.2.2 Assessment of sensitivity

As previously mentioned, the sensitivity factor consists of six main components. For the "water" component, the vulnerability index shows a value of 0.83 for HTU2 compared to 0.57 for HTU1. This difference is probably due to the significant discontinuity in water distribution. This leads to insufficient food supplies, generating conflicts between populations and revealing a new disparity between the two units. The same applies to "natural resources," which are characterized by a high index at HTU2 (0.76) compared to HTU1 (0.39). The main reason for this difference could be explained by the depletion of vegetation cover on steep slopes, which promotes various processes of water erosion and the overgazing. In terms of the "land tenure" component, this area remains vulnerable, with a ratio of 0.22 compared to 0.17 in HTU1. The latter unit has a higher proportion of households with areas of less than 5 ha and significant fragmentation problems. On average, we observe a division of farms by three for each owner, resulting in a high degree of fragmentation in the area. The "human resources" component also influences the sensitivity of households to climate change. In both territories, the proportion of women working on other farms is almost identical. Similarly, the rate of rural exodus is comparable in both areas, suggesting that out-migration affects households upstream and downstream alike.

3.2.3 Assessment of exposure

Focusing on the exposure factor, the component of climate variability and change leads to the same conclusion about the heightened vulnerability of the upstream area (0.61 compared to 0.51 in HTU1). This can be explained by the increase in the frequency of exceptional years, whether dry or wet (flooding), in recent years. This observation seems plausible when referring to the DGRE observation series (Figure 2). Several households (77.5%) reported the destruction of their soil and water conservation techniques and crops during the flood of 5 September 2020, when rainfall reached 72 mm. The vulnerability to extreme climate events is also observed in HTU1. Furthermore, 17.5% of households in this unit reported having made changes to their crop types and farming practices in order to better protect their assets.

TABLE 3 LVI-IPCC value in HTU1 and HTU2.

Main component	Sub-component	Value of the main component Mt	
		HTU1	HTU2
Socio-demographic profile	Average age of household heads (year) Female-headed households (%) Households with more than 7 family members (%) Households with more than 4 dependents (%) Households with only one working member per family (%) Households with unemployed family members (%)	0.26	0.24
Livelihood strategies	Household dependent on agriculture as their main source of income (%) Households with a family member working in another region (%) Households owning irrigated land (%) Households relying exclusively on rain-fed agriculture (%)	0.6	0.35
Social networks	Households that have not been members of any organization or association (%) Households that have received assistance from friends, associations, communities or the state (%)	0.65	0.4
Infrastructure	Households with difficult access to paved roads and farm tracks at all times (%) Households with poor-quality telecommunications network (%)	0.26	0.42
Health	Average time to reach a health facility (in minutes) Health expenditure as a proportion of total expenditure (per month) (%) Households with members who have missed work or school due to illness in the last 3 months (%) Households with members suffering from a chronic illness (%)	0.51	0.43
Food	Household food satisfaction (%) Households unable to meet the food needs of their livestock (%) Households' dependent on their family farm for food (%)	0.76	0.69
Water availability and access	Households reporting water-related conflicts (%) Households without drinking water (%) Households without a constant and regular water supply (%) Average number of days without a regular water supply per month Households that do not use supplementary irrigation during periods of drought (%) Households without semi-buried cisterns (%)	0.57	0.83
Natural resources	Households with eroded land (%) Households reporting loss of vegetation cover (%) Households reporting degradation of rangelands (%)	0.39	0.76
Human resources	Women working on other farms (in the region or elsewhere) (%) Rural exodus (%)	0.31	0.32
Land tenure	Households with small farms (area less than 5) (%) Households with more than 3 plots (%) Average farm size (ha)	0.17	0.22
Climate variability and change	Number of floods in recent years Number of dry years in recent years Households reporting changes in crop types due to climate change (%) Households reporting a change in farming practices (%) Households whose soil conservation techniques were destroyed by the 2020 floods (%) Households reporting that the 2020 floods destroyed their crops (%)	0.51	0.61
LVI-IPCC		0.03	0.14

Moreover, we observed that these households have abandoned rain-fed cereal and almond cultivation in favor of fruit orchards and irrigated vegetable crops. The latter relies on water from wells and boreholes, the number of which is constantly increasing. Finally, we can conclude that HTU2 (upstream) is more vulnerable than HTU1 (downstream) especially in terms of natural resource availability and management.

4 Discussion

The findings of this study confirm the findings of previous vulnerability studies, which emphasise that the adverse effects of climate change are particularly pronounced on the already fragile livelihoods of rural populations in rural areas of Tunisia, as well as on the

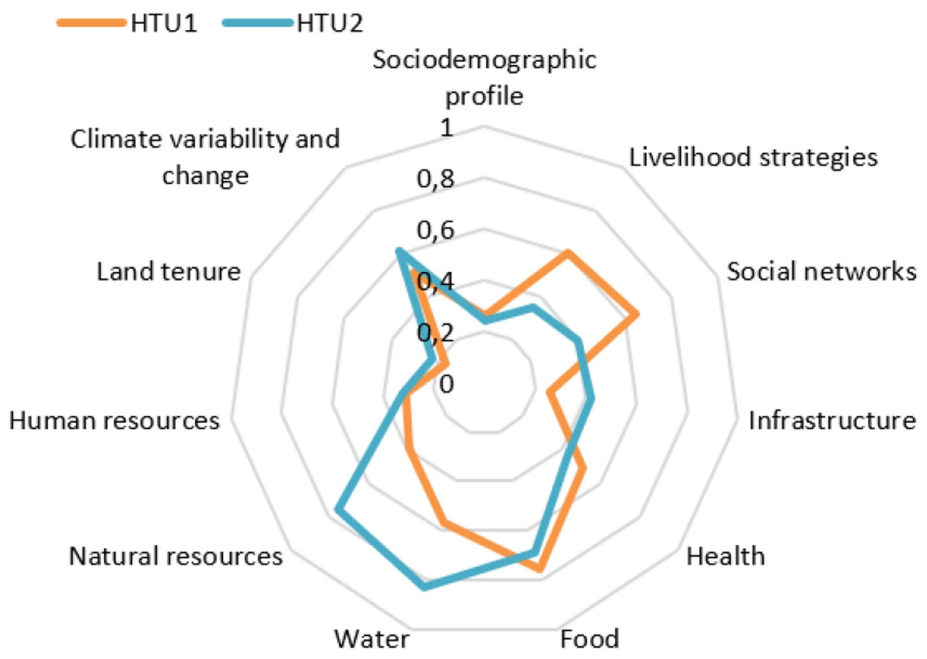

FIGURE 4
Vulnerability by main component.

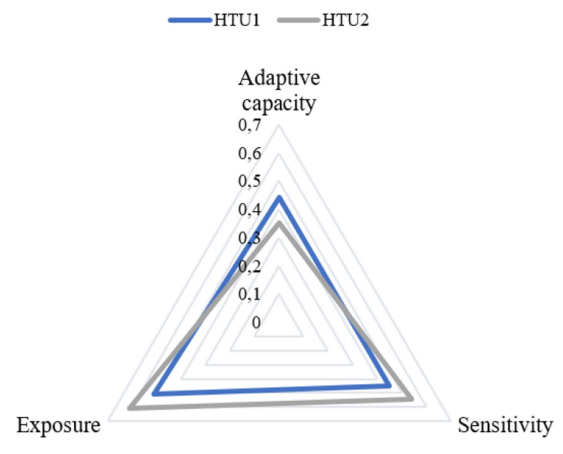

FIGURE 5
Vulnerability according to IPCC factors.

agroecosystems that constitute their living environment (World Bank, 2013; Ben Nasr et al., 2021; Aribi and Sghaier, 2021). However, this vulnerability varies significantly on a small scale. In our study, we found that the upstream part of Rihana (HTU2) is more vulnerable than the downstream part (HTU1) (LVI-IPCC of 0.14 for HTU2 and 0.03 for HTU1), which also confirms the perception of the local population. The results show that in the upstream part (HTU2), a significantly larger proportion of the population reports experiencing the main impacts, particularly the loss of fertility and soil erosion, the widening and deepening of the oueds, damage to water and soil conservation techniques, water scarcity, and the abandonment of certain agricultural lands, etc. In contrast, downstream participants (HTU1) recognized the same impacts but with lower frequency. This convergence between perceptions and quantitative measures shows that local populations have a detailed and realistic understanding of the vulnerability to climate change in their territories. This suggests that even on a small scale, some areas were more vulnerable than others. These differences can be explained by local conditions: position in the toposequence, diversity of available resources, agricultural practices, access to water and local infrastructure, as well as people's perceptions of climate change. These results are consistent with those of Fetoui et al. (2021), who emphasized that the vulnerability of rural areas in Tunisia is heterogeneous and strongly influenced by environmental and socio-economic factors. Similarly, Verlynde (2018), and Widayati et al. (2021) have shown that the perceptions of households and local communities play a key role in adaptive capacity and therefore in the level of vulnerability.

The perception of climate change by the population of Rihana shows that climate change is not just an idea but a lived reality. In HTU1, where agriculture is mainly rain-fed and highly dependent on natural conditions, households in this area attribute the changes observed to natural conditions. In contrast, in HTU2, where intensive and irrigated agriculture is expanding, households perceive that, beyond natural causes, human activities play an important role in the climate changes observed. This perception can be explained by their direct observation of the effects of human actions on local resources (deep and frequent ploughing, irrigation, massive use of chemical fertilisers) and easy access to information (awareness-raising, media discourse) highlighting the anthropogenic origin of climate change. In terms of the consequences of climate change, farmers in both HTUs identified that they had noticed, first and foremost, a change in rainfall patterns, followed by more frequent and prolonged droughts and temperature variations. With regard to sudden and violent floods, there is a notable difference between the two territories: farmers upstream, located in mountainous areas, observe these phenomena more frequently. These floods cause soil erosion, widening and deepening of the oueds and deterioration of soil conservation techniques. In addition, this population perceives that climate change can lead to a loss of production and a reduction in livestock farming, which in turn can

cause the abandonment of certain agricultural lands and rural exodus. For the population downstream, the main concern is the drying up or significant reduction in the flow of the oued, which is no longer able to supply their farms. This situation is largely attributed to the soil and water techniques used upstream, such as tabias, individual basins and Mgouds, which alter the natural flow of the watercourse. As a result, this pressure on water resources may also cause conflicts between upstream and downstream users. The results also show that households anticipate an increase in future vulnerability, with intensified soil erosion, declining rain-fed crops, growing water scarcity, and a heightened risk of rural exodus. These perceptions align with the findings of Chebbi et al. (2019), who project those areas suitable for cereal cultivation could decrease by an average 16% by 2,100, as well as a decline in cattle, and sheep, particularly in the South and Central of the country. In this context, agricultural households that rely on livestock farming will be the most vulnerable, as climate change will lead to land degradation, particularly through a reduction in grazing areas for livestock, which will have a negative impact on livestock productivity. As a result, the loss of grazing land is compelling livestock farmers to adapt by modifying their activities or relocating, alongside adjustments in agricultural land and water use (Elame et al., 2016).

The LVI-IPCC results indicate that HTU2 has high sensitivity and exposure, with a low adaptive capacity. This low adaptive capacity is mainly explained by the fact that the majority of households rely exclusively on agriculture as their main source of income, and that farming is predominantly rain-fed. In contrast, in HTU1, irrigated agriculture is practiced, especially after 2010, when irrigated farming expanded. These results confirm that exclusive dependence on rainfed agriculture increases the vulnerability of households, while irrigation is a key factor in resilience (Hahn et al., 2009; Adu et al., 2018). Another factor limiting the adaptive capacity of HTU2 is its infrastructure. Being a mountainous area, access to basic services in HTU2 is extremely challenging. The sensitivity livelihood vulnerability is mainly affected by water availability and access, and naturel resources constraints (Pandey and Jha, 2012; Mahdhi et al., 2019). Focusing in the exposure factor, findings show that the two territories are both exposed to recurring climatic disturbances, particularly floods and periods of drought that have become more frequent in recent decades. Yet, exposure remains slightly higher in the upstream area (0.61) compared with the downstream one (0.51). This difference is largely linked to the physical setting of the upstream landscape. Its uneven topography promotes rapid runoff and severe soil erosion, which amplifies flood impacts during heavy rainfall. The proximity of farmlands to oueds further increases flood risk, while shallow and fragile soils make them more vulnerable to drought and reduce their ability to retain water. Furthermore, reduced or degraded vegetation cover weakens the natural barriers that usually protect against erosion and surface water flows. The crop losses recorded during the 2020 floods, together with the observed shifts in cropping patterns and agricultural practices, clearly reflect this greater level of exposure. Overall, the results suggest that while both territories are vulnerable to climate change, the upstream area faces stronger physical and environmental pressures (FAO, 2018; Ben Salem et al., 2023).

The results also highlight that water scarcity is one of the most significant impacts of climate change in Rihana, a finding widely shared by the local population. Firstly, participants reported a marked decrease in winter rainfall, a predominance of light rainfall (less than 5 mm), an increase in temperature, and an intensification of prolonged and severe droughts, thus limiting groundwater recharge. This scarcity of resources is then reflected in a decrease in the area dedicated to rain-fed crops (olive trees and cereals), degradation of grazing land, and a change in traditional agricultural calendars, particularly in the upstream area. In contrast, in the downstream area, households are increasingly resorting to (often illegal) drilling, which has led to an expansion of irrigation since 2010. This trend has been encouraged by the presence of groundwater (at a depth of around 200 m), fertile land, easier access to credit thanks to land titles, and investment of income from emigration. The expansion of irrigation appears to be accentuating disparities between households located upstream and those downstream. This perspective corroborates the conclusions of Fautras (2021).

Beyond water scarcity, the population also highlights inequalities in access to resources between the two territories, upstream and downstream. Households in HTU2 mainly mentioned their suffering due to recurring hydrological deficits. Indeed, searching for water sources far from the douars is extremely difficult during dry seasons and years, without adequately meeting the needs of the inhabitants. The situation is even more serious when they mention the undersizing of the public drinking water distribution network. In addition, they spoke at length about illegal individual connections. These have caused a decrease in water flow, breakdowns in the pumping system, and also a drop in distribution pressure. As a result, water distribution is discontinuous, taking place in turns (only 4–5 days per month in summer and 10 h per day). This problem of undersizing is also present in the downstream section, but to a lesser extent than in HTU2. In addition, in this area, households use many semi-buried cisterns to store rainwater and purchased water in order to meet their various needs. The average purchase price of cisterns is 27 TND for 6 m^3, or 4.5 TND/m^3. This price varies between 15 and 40 DT/cistern depending on several criteria: access to the house (if access to the house is difficult, the cistern costs more), the distance between the source and the house, the quality of the purchased water, and the use of the cistern. Gana and Khemiri (2021) highlight this inequality of access to resources in rural areas of Tunisia, which reinforces the socio-spatial imbalance between upstream and downstream areas.

In the upstream area, heavy flooding combined with the relatively steep terrain causes erosion, leading to massive soil displacement. Another major factor contributing to erosion is the lack of dense vegetation cover capable of effectively holding the soil in place. As for the downstream section, it is marked by the expansion of areas dedicated to olive cultivation and the promotion of irrigation, which has played a key role in limiting the harmful effects of erosion. In addition, this area has 200 hectares of tabias, SWC developments created by the state in 2013 as part of the second national SWC strategy, promoting better soil conservation and reducing soil loss.

The vulnerability assessment was based on an original methodological framework combining two approaches: qualitative and quantitative. Experience confirms the value of integrating local knowledge with scientific knowledge to obtain a complete picture of vulnerability. It is important to involve the local population so that they can assess the vulnerability of their own territory for themselves. Group discussions, participatory mapping and participatory observations provide a structured basis for and support the results obtained from the LVI-IPCC calculation. Indeed, most authors focus solely on a quantitative analysis of vulnerability using indicators and maps (Madhuri, 2014; Fernández et al., 2015; Adu et al., 2017) without taking into account the population's perception of vulnerability in its territory (Tanguay and Viau, 2015). The combination of the two analyzes shows that the

characteristics of the local population's vulnerability are specific and variable depending on the climatic, physical, socio-economic and historical contexts, and result in a diversity of perceptions of climate change. The participatory vulnerability assessment, thanks to the participatory territorial diagnosis, made it possible to create a climate of trust with the local population and to acquire information that did not exist elsewhere. It made it possible to assess vulnerability in a more realistic way, very close to the reality on the ground and the vision of the local population. The adaptation of the LVI-IPCC index by integrating the components "water availability and access, infrastructure, natural resources, human resources and land tenure" was crucial for the implementation of the methodology in our study area. The choice of equal weighting was favored to ensure transparency and reproducibility of the calculations, an approach widely validated in the international literature (Hahn et al., 2009, Mwadzingeni et al., 2021; Kayastha et al., 2023; Zainab and Shah, 2024, etc.) and applied by numerous authors in Tunisia to assess rural and climate vulnerability (Aribi and Sghaier, 2021; Ben Nasr et al., 2021). The robustness of our results is based on a twofold comparative analysis: it not only identifies the most vulnerable area but also disaggregates the index to reveal which livelihood pillars are the most fragile. This approach highlights real vulnerability factors that remain consistent regardless of the aggregation scheme used. Thus, the index serves as a reliable basis for the prioritization of local actions.

The two approaches used have led to the acquisition and formalisation of knowledge not only at the HTU level but also encompassing more specific information within the study area. One example is the problem of collective land ownership, which generates conflicts between different members of the same family. This is also the case with gender-related issues. Women in Rihana are more sensitive to climate change and have weaker adaptive capacities than men. In addition, their livelihoods are weaker than those of men, with less access to resources (Shah et al., 2013). For example, during droughts, women find themselves in a vulnerable situation because they cannot afford to buy tanks to irrigate their plots. Several women stated that during the floods of 5 September 2020, they had neither the physical nor the financial means to maintain the water and soil conservation techniques that were destroyed by the floods. We also noted that in the participative exercises, women's participation is lower than that of men. The latter still seem to be dominant. This reality is well known in Sidi Bouzid and in the semi-arid and arid zones of Tunisia (Mazhoud and Ben Salem, 2025) and confirms the need to integrate the gender dimension into future studies on the issue of climate change adaptation.

5 Conclusion

Our study highlights the livelihood vulnerability of the local population of Rihana to climate change. The vulnerability assessment combined two complementary approaches, one qualitative and the other quantitative, to provide a comprehensive picture of the vulnerability of territories in Tunisia's semi-arid areas. The participatory territorial diagnosis analyzes the perceptions of the local population, while the vulnerability index based on the IPCC definition (LVI-IPCC), adapted to the semi-arid Tunisian context, was used to estimate vulnerability. The results of this study show, first, the importance of local population participation in assessing vulnerability in their territory. Second, the results highlight a variety of vulnerability situations observed at a very fine scale. These results confirm that the impacts of climate change manifest themselves in ways that are specific to each community, each of which faces unique challenges influenced by local factors: position in the toposequence, diversity of available resources, agricultural practices, access to water and local infrastructure, as well as people's perceptions of climate change. Thirdly, this study confirms the importance of using both approaches for the assessment of vulnerability. Local knowledge must be integrated with scientific knowledge to obtain a complete picture of vulnerability.

In conclusion, while climate change is a global issue, its consequences are highly localized. The impacts of climate change manifest themselves in ways that are specific to each community, with each facing unique challenges influenced by particular geographical, environmental, and socioeconomic factors. Rural communities in Tunisia face diverse challenges, and understanding the spatial and socio-environment variations in vulnerability is essential to building adaptive capacity and enhancing resilience at the local level. This study contributes to filling that knowledge gap, offering insights that are both scientifically rigorous and practically relevant for local stakeholders and policymakers. To effectively address these challenges, we suggest that the government, in implementing its adaptation policies, adopt a contextualized approach that takes into account local specificities by mobilizing a territorial approach and encouraging active participation by the local population. This approach will enable an effective response to highly local challenges and structured, participatory management of water resources.

The analysis of vulnerability to climate change based on the various types of territories will enable the mapping of vulnerability profiles over all of Tunisia's semi-arid zones. This ability to generalize is crucial for adaptation planning because it enables learning experiences in communities with similar vulnerability profiles and enables decision-makers to understand the trends or effects of a policy or program on the vulnerability of local communities' means of existence.

Finally, it is important to mention some limitations of our study. The sample size, which was 52 households, is the first limitation. It was not possible to include more due to constraints related to meetings organized during the post-COVID period. Another limitation is the participatory approach itself: although it is effective for gathering local perceptions and knowledge, it can introduce biases, such as the influence of certain dominant participants. Some people may minimize or exaggerate risks according to their interests. A third limitation concerns the use of equal weighting for all components of the index, which simplifies the calculation but may not reflect the actual relative importance of each component and subcomponent in the assessment of vulnerability.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

Ethical approval was not required for this study according to local regulations. All participants provided informed consent

before taking part in the surveys. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation was not required from the participants or the participants' legal guardians/next of kin in accordance with the national legislation and institutional requirements because Participation was voluntary, and all participants provided verbal informed consent in accordance with local practices.

Author contributions

NK: Writing – original draft, Writing – review & editing, Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Validation, Visualization. SJ: Conceptualization, Supervision, Validation, Writing – review & editing. NM: Methodology, Supervision, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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